

ISL-CENTRIC SIGN LANGUAGE TRANSLATION USING SPATIO-TEMPORAL KEYPOINT TRANSFER LEARNING

**Nomula Rishika
23951A66E4**

ISL-CENTRIC SIGN LANGUAGE TRANSLATION USING SPATIO-TEMPORAL KEYPOINT TRANSFER LEARNING

*A Project Report
submitted in partial fulfillment of the
requirements for the award of the degree of*

**Bachelor of Technology
in
CSE (Artificial Intelligence & Machine Learning)**

by

**Nomula Rishika
23951A66E4**



**Department of
CSE (Artificial Intelligence & Machine Learning)**

**INSTITUTE OF AERONAUTICAL ENGINEERING
(Autonomous)**

Dundigal, Hyderabad – 500 043, Telangana

January, 2026

© 2026, Nomula Rishika. All rights reserved.

DECLARATION

I certify that

- a. The work contained in this report is original and has been done by me under the guidance of my supervisor(s).
- b. The work has not been submitted to any other Institute for any degree or diploma.
- c. I have followed the guidelines provided by the Institute for preparing the report.
- d. I have confirmed the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- e. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Place:

Date:

Signature of the Student

Roll No.

CERTIFICATE

This is to certify that the project report entitled **ISL-Centric Sign Language Translation Using Spatio-Temporal Keypoint Transfer Learning** submitted by **Ms Nomula Rishika** of Aeronautical Engineering, Hyderabad in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in **CSE (Artificial Intelligence & Machine Learning)** is a Bonafide record of work carried out by him/her under my guidance and supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute for the award of any Degree.

Supervisor

Head of the Department

Date:

APPROVAL SHEET

This project report entitled **ISL-Centric Sign Language Translation Using Spatio-Temporal Keypoint Transfer Learning** by **Nomula Rishika** is approved for the award of the Degree Bachelor of Technology in **CSE (Artificial Intelligence & Machine Learning)**.

Examiners

Supervisor

Principal

Date:

Place:

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without acknowledging the people who made it possible and whose constant guidance and encouragement crowned all efforts with success.

I express my sincere gratitude and profound indebtedness to my project guide, **Ms. B. Shirisha, Assistant Professor**, Department of Computer Science and Engineering (Artificial Intelligence & Machine Learning), for his valuable guidance, constant support, and constructive suggestions throughout the course of this project work.

I express my sincere gratitude to **Mr. K Subba Shankar, Assistant Professor**, Project Coordinator, for his effective coordination, constant encouragement, and timely guidance throughout the course of this project. His valuable suggestions, systematic monitoring, and readiness to clarify doubt played a crucial role in the smooth progress and successful completion of this work.

I am grateful to **Dr. M. Purushotham Reddy**, Professor and Head of the Department, Department of **Computer Science and Engineering (Artificial Intelligence & Machine Learning)**, for his encouragement and support in carrying out this project successfully.

I express my sincere thanks to **Dr. L. V. Narasimha Prasad**, Professor and **Principal**, for being a great source of inspiration and for providing the necessary facilities to accomplish this work.

I also thank the college management and **Sri M. Rajashekar Reddy**, Chairman of the Institute of Aeronautical Engineering (IARE), Dundigal, for providing the necessary infrastructure and resources to complete this project.

Finally, I take this opportunity to express my heartfelt gratitude to all those who directly or indirectly supported and helped me in bringing this project to its present form.

ABSTRACT

Keywords: Indian Sign Language recognition, hand keypoint extraction, MediaPipe-based gesture analysis, deep learning, transfer learning, and real-time sign language translation.

Indian Sign Language (ISL) recognition plays a crucial role in improving communication accessibility for the deaf and hard-of-hearing community. In the initial phase of this project, an ISL sentence-level video dataset was used to train a gesture recognition model; however, due to the limited number of samples per sentence, splitting the dataset into training and testing sets resulted in poor generalization, achieving an accuracy of only 41%, which highlighted the difficulty of learning robust representations from small datasets. To address this limitation, the project was redesigned using a larger and more structured ISL dataset consisting of numerical (0–9) and alphabetic (A–Z) gestures. Hand keypoints were extracted from images using MediaPipe, transforming each gesture into a compact 63-dimensional feature representation. A deep learning model trained on these keypoints achieved approximately 99% accuracy, demonstrating strong and consistent class-wise performance across both numbers and alphabets. Furthermore, an ISL-centric transfer learning approach was explored by pretraining a base model on numerical gestures and fine-tuning it for alphabet recognition; experimental results showed that while transfer learning is effective in low-data scenarios, training from scratch outperformed transfer learning when sufficient target-domain data was available. Finally, the trained model was integrated with a webcam to enable real-time ISL gesture recognition and live text translation, confirming the effectiveness of keypoint-based deep learning for accurate and real-time ISL recognition.

CONTENTS

Title Page	I
Declaration	II
Certificate by the Supervisor	III
Approval Sheet	IV
Acknowledgement	V
Abstract	VI
Contents	VII
List of Tables	VIII
List of Figures	IX
List of Abbreviations	X
Abstract	XI
Chapter 1 Introduction	1-9
1.1 Introduction	1-2
1.2 Objectives	2 -3
1.3 Feasability	3
1.4 Existing Methodologies	3-5
1.5 Demerits of the existing methologies	5-7
1.6 System Requirements	8-10
Chapter 2 Review of Relevant Literature	11-18
Chapter 3 Methodology	19-23
Chapter 4 Results and Discussions	24-31
Chapter 5 Conclusions and Future Scope of Study	32-34
References	35-37

LIST OF TABLES

Table No	Name of the Table	Page No.
4.1	Comparative Analysis	28

LIST OF FIGURES

Figure No.	Name of the Figure	Page No.
3.1	System Architecture	20
3.2	Labelled Keypoints	21
3.3	Model Architecture	22
4.1	ISL hand gesture images representing numbers(0-9)	24
4.2	Train and Test Accuracy	26
4.3	Transfer Learning Results	27
4.4	ISL Live Translation for alphabet B	29
4.5	ISL Live Translation for alphabet O	29
4.6	ISL Live Translation for Number 3	30
4.7	Confusion Matrix	31

LIST OF ABBREVIATIONS

ABBREVIATIONS	DEFINITION
ISL	Indian Sign Language
ASL	American Sign Language
SLR	Sign Language Recognition
CNN	Convolutional Neural Networks
LSTM	Long Short Term Memory
GCN	Graph Convolutional Networks
RNN	Recurrent Neural Networks
HMM	Hidden Markov Model

CHAPTER 1

INTRODUCTION

1.1 Introduction

Sign language serves as a fundamental mode of communication for the deaf and hard-of-hearing community, enabling them to express thoughts, emotions, and information through structured hand gestures, movements, and facial expressions. In India, Indian Sign Language (ISL) is widely used; however, effective communication between ISL users and the hearing population remains a major challenge due to the lack of widespread awareness and technological support. This communication barrier often limits access to education, employment, and public services for the deaf community.

Recent advancements in computer vision and artificial intelligence have created opportunities to develop automated sign language recognition systems. Traditional approaches to gesture recognition relied heavily on sensor-based devices such as data gloves, which, although accurate, were expensive and impractical for everyday use. Vision-based approaches using raw images and videos emerged as a more accessible alternative but were often affected by variations in lighting, background clutter, camera quality, and user hand orientation.

To overcome these limitations, modern sign language recognition systems increasingly employ hand keypoint-based techniques. By focusing on the geometric structure of the hand rather than raw pixel values, keypoint-based representations significantly reduce data dimensionality and improve robustness against environmental variations. In this project, MediaPipe is used to extract 21 hand landmarks, each represented by three-dimensional coordinates, resulting in a compact and informative feature vector suitable for deep learning models.

This project initially explored sentence-level ISL recognition using video datasets; however, the limited number of samples per sentence led to poor model generalization and low accuracy. Recognizing this challenge, the approach was redesigned using a larger and more structured dataset consisting of ISL numerical and alphabetic gestures. A deep learning model trained on the extracted hand keypoints achieved high

recognition accuracy, demonstrating the effectiveness of the proposed approach. Furthermore, the system was integrated with a webcam to enable real-time gesture recognition and live text translation, highlighting its potential for practical deployment in real-world communication scenarios.

In addition to accuracy, real-time performance is a critical requirement for practical sign language recognition systems. A system that performs well only in offline testing but fails under real-world conditions such as varying lighting, hand orientations, and motion cannot be considered effective. Therefore, this project emphasizes not only high classification accuracy but also real-time usability by integrating the trained model with a live webcam interface. By combining efficient hand keypoint extraction with lightweight deep learning models, the proposed system ensures fast inference and stable predictions, making it suitable for real-time ISL gesture recognition. This approach demonstrates how modern computer vision techniques can be applied to develop accessible and user-friendly assistive technologies.

1.2 Objectives

The Indian Sign Language (ISL) recognition mechanism proposed using hand keypoint-based deep learning techniques incorporates the following main goals:

- **Real-Time ISL Gesture Recognition:** The primary objective is to recognize Indian Sign Language gestures in real time using webcam input in order to enable effective and instant communication between ISL users and non-signers.
- **High-Accuracy Gesture Classification Using Keypoint-Based Learning:** To improve recognition accuracy by extracting hand keypoints using MediaPipe and training deep learning models capable of learning spatial patterns in ISL numerical and alphabetic gestures.
- **Efficient Feature Representation and Model Performance:** To reduce the influence of background noise, lighting variations, and camera inconsistencies by utilizing compact hand landmark representations instead of raw image data.
- **Exploration of Transfer Learning for ISL Recognition:** To investigate the effectiveness of ISL-centric transfer learning by reusing learned representations from numerical gestures for alphabet recognition and analyzing its impact on

classification performance.

1.3 Feasibility

The proposed Indian Sign Language (ISL) recognition system is feasible from technical, economic, and operational perspectives. From a technical standpoint, the system is developed using open-source technologies such as Python, MediaPipe, TensorFlow, OpenCV, and NumPy, which are widely supported and well-documented. Hand keypoint extraction significantly reduces computational complexity compared to image-based approaches, allowing the model to run efficiently on standard computing systems without the need for specialized hardware. The availability of a large and well-structured ISL dataset further ensures the feasibility of training accurate and reliable models.

From an economic perspective, the project is cost-effective as it does not require expensive sensors or proprietary software. All tools, libraries, and datasets used in the system are freely available, minimizing development and deployment costs. The system can be executed on commonly available devices such as laptops or desktop computers equipped with a basic webcam, making it accessible for academic and real-world use.

Operationally, the system is user-friendly and easy to deploy. The webcam-based interface allows users to perform ISL gestures naturally without additional equipment or training. The modular design of the system enables future enhancements, such as sentence-level recognition, speech output, and support for additional sign languages, without major changes to the existing architecture. Overall, the project is feasible, scalable, and suitable for practical implementation.

1.4 Existing Methodologies

Sign language recognition has been an active area of research for several years, and various methodologies have been proposed to interpret hand gestures and translate them into meaningful text or speech. These methodologies can broadly be classified into sensor-based approaches and vision-based approaches.

1.4.1. Sensor-Based Methodologies

Sensor-based systems were among the earliest approaches for sign language recognition. These systems use specialized devices such as data gloves, flex sensors, accelerometers, and motion trackers to capture finger bending, hand orientation, and movement dynamics. The captured

sensor data is then processed using machine learning algorithms for gesture classification. Although sensor-based approaches provide precise measurements and high recognition accuracy, they suffer from major drawbacks such as high cost, limited user comfort, calibration requirements, and dependency on external hardware. These limitations restrict their usability in real-world and large-scale applications.

1.4.2 Vision-Based Image Processing Techniques

Vision-based systems eliminate the need for wearable devices by using cameras to capture hand gestures. Early vision-based methodologies relied heavily on traditional image processing techniques such as thresholding, skin color detection, contour extraction, and morphological operations. Handcrafted features including shape descriptors and texture features were extracted and classified using traditional classifiers like Support Vector Machines (SVMs) and k-Nearest Neighbors (k-NN). While these approaches reduced hardware complexity, they were highly sensitive to environmental conditions such as lighting variation, complex backgrounds, and camera resolution, leading to inconsistent performance.

1.4.3 Deep Learning-Based Vision Approaches

With the rise of deep learning, convolutional neural networks (CNNs) became the dominant approach for sign language recognition using images and videos. CNNs automatically learn hierarchical spatial features from raw image data, improving recognition accuracy over handcrafted methods. For dynamic sign recognition involving continuous gestures or sentences, recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks are used to capture temporal dependencies across video frames. Despite their effectiveness, deep learning-based vision models require large annotated datasets and significant computational resources, making them challenging to train and deploy for real-time applications.

1.4.4 Keypoint-Based Recognition Methods

Recent advancements have introduced keypoint-based methodologies that focus on extracting skeletal or landmark representations of the hand. Frameworks such as MediaPipe detect hand landmarks and represent gestures using joint coordinates, significantly reducing input dimensionality while preserving essential structural information. These approaches offer improved robustness against lighting changes, background noise, and camera variations.

Keypoint-based systems are computationally efficient, require less training data compared to image-based models, and are well-suited for real-time gesture recognition.

1.4.5 Transfer Learning in Sign Language Recognition

Transfer learning has been explored to address data scarcity by reusing knowledge from pretrained models. In sign language recognition, models trained on one set of gestures or domains are fine-tuned for related tasks. While transfer learning is beneficial when target datasets are small, its effectiveness decreases when sufficient task-specific data is available. In some cases, training a model from scratch yields better performance, highlighting the importance of dataset size and domain similarity.

1.5 Demerits of Existing System

Despite significant advancements in sign language recognition systems, existing approaches still face several technical, practical, and deployment-related limitations. These challenges restrict their effectiveness, scalability, and real-world usability, particularly for Indian Sign Language (ISL) applications.

1.Limited Dataset Availability

One of the most critical limitations is the lack of large, diverse, and well-annotated sign language datasets, especially for ISL. Many existing datasets contain a small number of samples, limited sign vocabulary, or insufficient variations in signers, lighting conditions, and camera angles. When such datasets are split into training and testing sets, the reduced sample size often leads to poor generalization and overfitting, resulting in low accuracy, as observed in earlier sentence-level ISL recognition attempts.

2. Sensitivity to Environmental Conditions

Vision-based sign language recognition systems are highly sensitive to external factors such as lighting variations, background clutter, camera quality, and occlusions. Traditional image-processing-based methods are particularly affected, as they rely on pixel-level features. Even deep learning-based image models can struggle when gestures are performed under different lighting or background conditions than those seen during training.

3. High Computational Complexity

Many deep learning models, especially those processing raw video frames, require significant computational resources for training and inference. Models combining CNNs with LSTMs or Transformers for temporal modeling often demand powerful GPUs and high memory usage. This makes real-time implementation challenging on low-resource devices and limits the deployment of such systems in practical environments.

4. Dependency on Specialized Hardware

Sensor-based systems rely on wearable devices such as data gloves or motion sensors, which increase system cost and reduce user comfort. These systems also require calibration and maintenance, making them impractical for everyday use. The need for specialized hardware restricts scalability and widespread adoption.

5. Poor Generalization Across Users

Many existing systems perform well only for the users included in the training data. Variations in hand size, signing speed, signing style, and gesture articulation among different users lead to reduced accuracy when the system encounters unseen signers. This user dependency is a major obstacle in building inclusive and robust sign language translation systems.

6. Inadequate Handling of Dynamic Gestures

While several systems focus on static hand gestures, real-world sign language consists of dynamic, continuous gestures involving temporal transitions, facial expressions, and body movements. Existing approaches often struggle to accurately segment and recognize continuous signs, leading to incorrect or fragmented translations, especially at the sentence level.

7. Difficulty in Sentence-Level Translation

Most existing systems are designed for isolated word or character recognition. Extending these systems to sentence-level translation introduces challenges such as temporal alignment, coarticulation effects, and contextual understanding. Limited training data and complex sequence modeling further reduce performance, as evidenced by low accuracy in

sentence-level ISL recognition tasks.

8. Limited Real-Time Performance

Achieving both high accuracy and low latency remains a challenge. Systems that achieve high accuracy often suffer from delayed response times, making them unsuitable for real-time communication. Conversely, systems optimized for speed may compromise recognition accuracy.

9. Lack of Standardization and Evaluation Metrics

There is no widely accepted benchmark dataset or standardized evaluation protocol for ISL recognition. Different studies use different datasets, preprocessing techniques, and metrics, making fair comparison and reproducibility difficult.

1.6 System Requirements

To build, train, and deploy a multi-class classification model for Indian Sign Language (ISL) recognition using hand keypoints and deep learning, the following system requirements should be considered:

Hardware Requirements:

The successful training, development, and deployment of DecepTrack: Improving Fake News Detection using Neural Fusion necessitate a computing platform that is high performance but also highly accessible. The use of LSTM as the underlying model guarantees that the hardware requirements are moderate relative to complex architectures like transformers or ensembles. The following is the breakdown of the recommended and minimum hardware specifications:

1. Processor (CPU)

- **Minimum:** Intel Core i5 (2.0 GHz or more, 4 cores)
- **Recommended:** Intel Core i7 / AMD Ryzen 7 or above
- The CPU is responsible for video capture, data preprocessing, keypoint extraction, and inference computations. Multi-core CPUs enhance performance, especially when processing multiple video frames simultaneously.

2. Memory (RAM)

- **Minimum:** 8 GB
- **Recommended:** 16 GB or higher
- RAM is used to cache input sequences, hand keypoint arrays, and intermediate training variables. Sufficient memory is important when processing large datasets or performing real-time gesture recognition to avoid heavy disk usage.

3. Storage

- **Minimum:** 5 GB free disk space
- **Recommended:** SSD with 20 GB or higher
- Storage is required for datasets (images or keypoint .npy files), trained models, checkpoints, logs, and outputs. SSDs ensure faster read/write operations during training and inference.

4. Graphics Processing Unit (GPU) – Optional but Recommended

- **Recommended:** NVIDIA RTX 2060 / 3060 (6+ GB VRAM) or higher with CUDA support
- Although the system can run on CPU, GPU accelerates training of neural networks, especially for transfer learning, and improves real-time recognition performance. Free cloud GPU platforms such as Google Colab or Kaggle Kernels can also be used.

5. Webcam

- HD (720p) or Full HD (1080p) USB webcam
- Required for capturing live hand gestures for real-time demonstration and testing.

6. Monitor

- 1080p resolution or higher
- To visualize gesture recognition outputs, keypoint overlays, and evaluation metrics such as confusion matrices.

Software Requirements:

For smooth development, training, testing, and deployment of the ISL recognition system, a stable software environment is required. The software stack is open-source and deployable both locally and on cloud platforms. Required components include:

1. Operating System

- **Minimum Requirement:** Windows 10 / Ubuntu 18.04 LTS or later
- **Recommended:** Ubuntu 20.04 LTS or later
- Linux-based environments offer better compatibility with Python ML libraries and GPU support.

2. Programming Language

- **Primary Language:** Python 3.9 or above
- Python is chosen for its simplicity, extensive library support, and vibrant ML community, enabling rapid prototyping and integration of deep learning models.

3. Python Libraries and Frameworks

- **TensorFlow 2.x / Keras:** For constructing and training the neural network models

- **MediaPipe:** For extracting hand keypoints
- **OpenCV:** For video capture and image preprocessing
- **NumPy / Pandas:** Numerical computation and dataset manipulation
- **Matplotlib / Seaborn:** Plotting and visualization of metrics such as confusion matrices, accuracy, and loss curves
- **scikit-learn:** Dataset splitting, evaluation metrics, and other ML utilities

4. Development Tools

- **IDE:** Visual Studio Code / PyCharm / JupyterLab
- **Version Control:** Git for code versioning and collaboration
- **Environment Management:** Anaconda / Miniconda (optional, for dependency isolation)

5. Optional Software (for Deployment)

- **Flask / FastAPI:** To serve the model through a REST API
- **Docker:** For containerizing the application for portability
- **Postman:** For testing API endpoints
- **Google Colab / Kaggle Kernels:** Cloud-based GPU training
- **GitHub / GitLab:** Code hosting and project version control
- **Streamlit / Hugging Face Spaces:** For creating interactive web applications to demo the ISL recognition system

CHAPTER 2

LITERATURE REVIEW

Zhang F. et al. (2021)[1] – Skeleton-Based Hand Gesture Recognition Using MediaPipe

Methodology:

This study employed MediaPipe to extract hand and pose keypoints and used these skeletal features for gesture classification. By focusing on keypoints instead of raw images, the system reduced background noise and lighting dependency.

Drawbacks:

The work focused mainly on static hand gestures and limited gesture sets. Temporal dynamics and sentence-level recognition were not explored. The study also did not compare transfer learning with training-from-scratch models.

Ong S.C.W., Ranganath S. (2005) [2]– Automatic Sign Language Analysis: A Survey

Methodology:

This survey paper analyzed various sign language recognition techniques, including vision-based, glove-based, and hybrid systems. It compared feature extraction methods, classification techniques, and datasets used in early research.

Drawbacks:

While comprehensive, the study did not propose a practical implementation. Most techniques discussed were outdated and relied on limited datasets. The survey lacked focus on real-time systems and emerging deep learning approaches.

Pigou L. et al. (2018) [3]– Beyond Temporal Pooling: Recurrence and Temporal Convolutions for Gesture Recognition

Methodology:

This work utilized Convolutional Neural Networks combined with temporal modeling techniques to recognize gestures from video data. The approach extracted spatial features from video frames and captured temporal dependencies using recurrent layers.

Drawbacks:

The model required large annotated video datasets and high computational resources. It was primarily evaluated on controlled datasets and did not address Indian Sign Language or real-

time deployment challenges.

Camgoz N.C. et al. (2020)[4] – Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation

Methodology:

This paper proposed an end-to-end deep learning architecture using transformer models to perform sign language recognition and translation directly from video sequences. The model jointly learned spatial and temporal representations.

Drawbacks:

The approach was highly data-intensive and computationally expensive. It required large-scale datasets that are not available for ISL. The complexity of the model makes real-time deployment difficult on low-resource systems.

Starner T., Pentland A. (1997)[5] – Real-Time American Sign Language Recognition Using Desk and Wearable Computers

Methodology:

This study introduced one of the earliest real-time sign language recognition systems using Hidden Markov Models (HMMs). Video sequences were processed to extract hand motion features, which were then modeled temporally using HMMs. The system focused on continuous gesture recognition and laid the foundation for sequence-based sign recognition.

Drawbacks:

The system relied heavily on handcrafted features and controlled environments. It was sensitive to lighting conditions and background variations. Additionally, the computational cost was high, and the approach did not generalize well to different signers or real-world scenarios.

Keskin C., Kırac F., Kara Y.E., Akarun L. (2013) [6]– Hand Pose Estimation and Hand Shape Classification Using Multi-layered Randomized Decision Forests

Methodology:

This paper proposed a real-time hand pose estimation system using depth sensors. Hand joint positions were extracted and classified using randomized decision forests. The system was designed to work efficiently for hand tracking and pose recognition.

Drawbacks:

The approach required specialized depth sensors, increasing system cost. It was sensitive to sensor noise and limited to controlled environments. The method did not address full-body gestures or sign language sentence construction.

Neverova N., Wolf C., Taylor G., Nebout F. (2014)[7] – Multi-scale Deep Learning for Gesture Detection and Localization

Methodology:

This study applied multi-scale convolutional neural networks to detect and localize gestures from video sequences. Spatial and temporal features were jointly learned using deep learning models.

Drawbacks:

The system required high computational power and large annotated datasets. It focused on gesture detection rather than semantic interpretation, making it unsuitable for sign language translation tasks.

Molchanov P., Gupta S., Kim K., Kautz J. (2016)[8] – Hand Gesture Recognition with 3D Convolutional Neural Networks

Methodology:

This work utilized 3D CNNs to capture spatio-temporal features from gesture videos. The model processed video volumes directly, enabling dynamic gesture recognition.

Drawbacks:

Processing raw video volumes resulted in high memory and computation requirements. The approach was unsuitable for real-time applications on low-resource systems and did not address ISL.

Koller O., Zargaran S., Ney H., Bowden R. (2017)[9] – Deep Sign: Hybrid CNN-HMM for Continuous Sign Language Recognition

Methodology:

This study combined CNN-based feature extraction with HMM-based temporal modeling for continuous sign language recognition. The hybrid architecture aimed to model both spatial and temporal aspects of signing.

Drawbacks:

The hybrid architecture increased system complexity and required careful tuning. Performance depended heavily on large labeled datasets, which are scarce for Indian Sign Language.

Simonyan K., Zisserman A. (2014)[10] – Two-Stream Convolutional Networks for Action Recognition in Videos

Methodology:

The two-stream CNN architecture processed spatial and temporal information separately using RGB frames and optical flow. The approach was influential in video-based action recognition tasks.

Drawbacks:

The method relied on optical flow computation, which is computationally expensive. It was not optimized for real-time sign language recognition and did not focus on fine-grained hand movements.

Ronchetti F., Quiroga F., Estrella J.C. (2016) [11]– Sign Language Recognition Without Vision-Based Sensors

Methodology:

This study explored wearable sensor-based sign language recognition using accelerometers and gyroscopes. Signals were processed and classified using machine learning techniques.

Drawbacks:

The requirement for wearable devices reduced user convenience. The approach lacked visual context and was impractical for public or spontaneous communication.

Abid A., Hussain F., Zafar A. (2019) [12]– Real-Time Sign Language Recognition Using MediaPipe and Deep Learning

Methodology:

This work used MediaPipe for hand landmark extraction and trained deep neural networks for sign classification. The system achieved improved robustness against background noise.

Drawbacks:

The study focused on a small gesture set and did not explore sentence-level recognition. Evaluation was limited, and comparisons with transfer learning approaches were not included.

Camgoz N.C., Koller O., Hadfield S., Bowden R. (2018) [13]– Neural Sign Language Translation

Methodology:

This research introduced an encoder-decoder neural network for translating sign language videos into text sentences. The model jointly learned recognition and translation.

Drawbacks:

The system required very large annotated datasets and powerful hardware. It was not suitable for real-time deployment and did not support ISL.

Liu J., Wang G., Hu W. (2020)[14] – Skeleton-Based Action Recognition Using Graph Convolutional Networks

Methodology:

This study modeled human skeletal data as graphs and applied Graph Convolutional Networks (GCNs) to recognize actions based on joint relationships.

Drawbacks:

While effective for action recognition, the approach was not specifically designed for sign language. Fine-grained hand articulation was not adequately modeled.

Shrestha A., Mahmood A. (2019) [15]– Review of Deep Learning Algorithms and Architectures

Methodology:

This survey reviewed various deep learning architectures used in computer vision and sequence modeling, including CNNs, RNNs, and Transformers.

Drawbacks:

The paper was theoretical and did not focus on practical sign language recognition systems or real-time constraints.

He K., Zhang X., Ren S., Sun J. (2016) [16] – Deep Residual Learning for Image Recognition

Methodology:

This work introduced Residual Networks (ResNet), which use skip connections to enable very

deep neural networks to be trained effectively. The residual learning framework improves feature extraction and has been widely adopted in vision-based gesture and sign language recognition systems.

Drawbacks:

Although highly accurate, ResNet models are computationally intensive and require optimization for real-time applications. The architecture focuses on spatial features and does not inherently capture temporal dynamics needed for continuous sign language recognition.

Graves A., Fernández S., Gomez F., Schmidhuber J. (2006) [17] – Connectionist Temporal Classification (CTC)

Methodology:

CTC is a sequence labeling technique designed for unsegmented temporal data. It enables RNNs to learn alignments between input sequences and output labels, making it effective for speech and continuous sign language recognition.

Drawbacks:

CTC training can be unstable and requires large amounts of labeled sequence data. It does not explicitly model spatial features such as hand keypoints or pose information.

Cooper H., Holt B., Bowden R. (2011) [18] – Sign Language Recognition

Methodology:

This chapter presents an overview of sign language recognition techniques, including hand tracking, feature extraction, and classification methods. It discusses both isolated and continuous sign recognition systems.

Drawbacks:

The methods discussed are largely based on traditional computer vision techniques. Deep learning-based approaches and real-time systems are not extensively covered due to the publication period.

Cao Z., Hidalgo G., Simon T., Wei S.-E., Sheikh Y. (2021) [19] – OpenPose: Realtime Multi-Person 2D Pose Estimation

Methodology:

OpenPose uses part affinity fields to detect body, hand, and facial keypoints in real time. It

enables accurate skeletal feature extraction, which is highly useful for gesture and sign language recognition tasks.

Drawbacks:

The framework is computationally expensive and may struggle on low-resource devices. Hand keypoint accuracy can degrade under occlusion or poor lighting conditions.

Rastgoo R., Kiani K., Escalera S. (2018) [20] – Hand Sign Language Recognition Using Deep Learning: An Overview

Methodology:

This work provides a comprehensive survey of deep learning techniques applied to hand sign language recognition, covering CNNs, RNNs, and hybrid architectures. It analyzes vision-based and sensor-based approaches, datasets, and evaluation strategies used in sign language research.

Drawbacks:

Being a survey paper, it does not propose a novel model or real-time implementation. Performance comparisons depend on previously published results and may not reflect recent advances.

2.3 Limitations of Existing Systems

From the reviewed literature, several limitations are evident:

- Heavy reliance on **raw image or video data**, leading to sensitivity to background and lighting.
- **Limited focus on Indian Sign Language**, with most studies centered on ASL or other Western sign languages.
- **Small and imbalanced datasets**, reducing generalization capability.
- Lack of **real-time implementation and evaluation**.
- Minimal exploration of **keypoint-based transfer learning**.
- Insufficient use of **comprehensive evaluation metrics** beyond accuracy.

2.4 Research Gap Analysis

Based on the literature review, the following research gaps have been identified:

1. Absence of robust **ISL-specific recognition systems** using skeletal keypoints.

2. Limited studies comparing **from-scratch learning and transfer learning** for sign language recognition.
3. Inadequate real-time systems suitable for **low-compute environments**.
4. Lack of **sentence-level output construction** from continuous predictions.
5. Insufficient evaluation using metrics such as **confusion matrix, precision, recall, and F1-score**.

2.5 Motivation for the Proposed System

The proposed system aims to bridge the identified gaps by using MediaPipe-based keypoint extraction combined with deep learning classifiers. It focuses on ISL alphabets and numbers, supports real-time recognition, and includes a comparative study of transfer learning and training-from-scratch approaches. The system also emphasizes robust evaluation and practical usability.

CHAPTER 3

METHODOLOGY

3.1 Overview of the Proposed System

The proposed system aims to develop an efficient and real-time Indian Sign Language (ISL) recognition and translation framework using hand keypoint extraction and deep learning techniques. The methodology focuses on extracting spatial hand landmarks instead of processing raw video frames, thereby reducing computational complexity while preserving essential gesture information. The system is designed to recognize ISL alphabets and numbers and translate them into readable text output.

Initially, a sentence-level ISL video dataset was explored; however, due to its limited sample size, the trained model achieved low accuracy. To overcome this limitation, a larger and more structured ISL dataset containing individual alphabet and number gestures was adopted. The proposed methodology incorporates feature extraction using MediaPipe, classification using deep neural networks, and real-time deployment using webcam-based gesture recognition.

3.2 System Architecture

The overall system architecture consists of the following major components:

1. **Data Acquisition**
2. **Keypoint Extraction**
3. **Feature Representation**
4. **Dataset Labeling**
5. **Model Training and Optimization**
6. **Transfer Learning Strategy**
7. **Real-Time Gesture Recognition and Translation**

Each module is designed to operate independently while contributing to the overall pipeline.

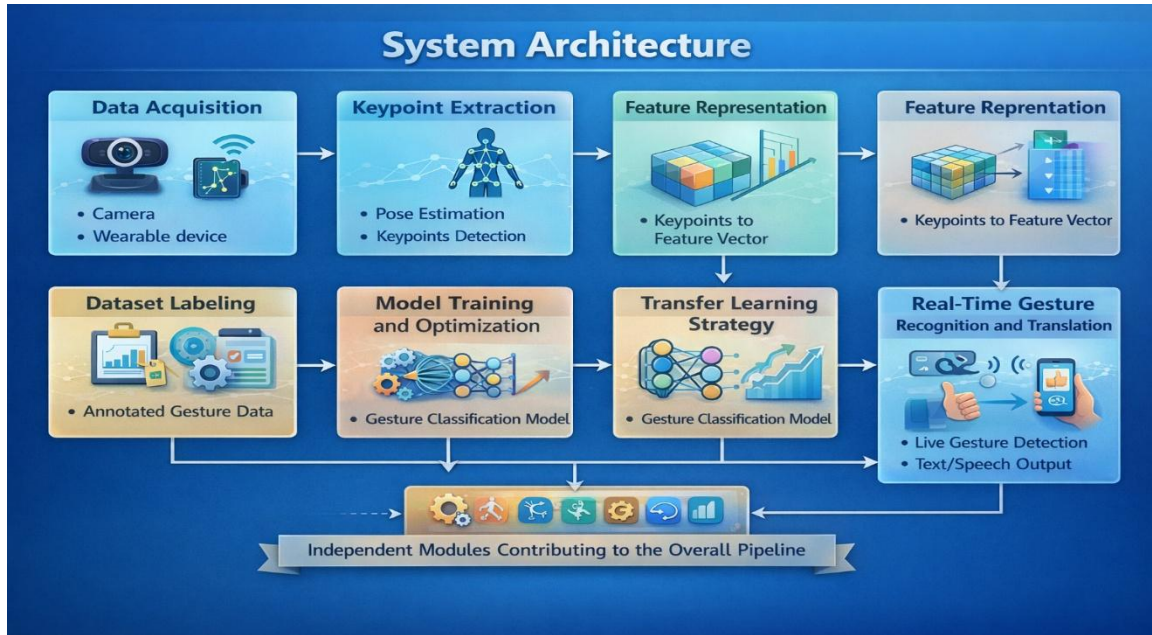


Figure 3.1 : System Architecture

Figure 3.1 : The System Architecture presents the complete system architecture for a gesture recognition pipeline. It starts with data acquisition using cameras or wearable devices, followed by keypoint extraction through pose estimation. These keypoints are converted into feature vectors and labeled to form a structured dataset. The data is then used for model training and optimization, enhanced with a transfer learning strategy. Finally, the trained model enables real-time gesture recognition and translation into text or speech, with each module working independently within the pipeline.

3.3 Data Acquisition

The dataset used in this project consists of static ISL hand gesture images representing numbers (0–9) and alphabets (A–Z). Each class contains approximately 1200 samples, with minor class imbalance in a few alphabet categories. The dataset images are stored in class-wise directories and are captured under varying lighting and background conditions to improve generalization. To enhance dataset usability, all images are preprocessed to ensure consistency in resolution and format before feature extraction

3.4 Keypoint Extraction Using MediaPipe

MediaPipe Hands is employed to extract 21 hand landmarks from each input image. Each landmark contains three-dimensional coordinates (x, y, z), resulting in a total of 63 features per image. This keypoint-based representation captures the geometric structure of hand gestures effectively.

Key advantages of this approach include:

- Reduction in computational complexity
- Robustness to background noise
- Independence from color and lighting variations

The extracted keypoints are saved as NumPy arrays and used as input features for model

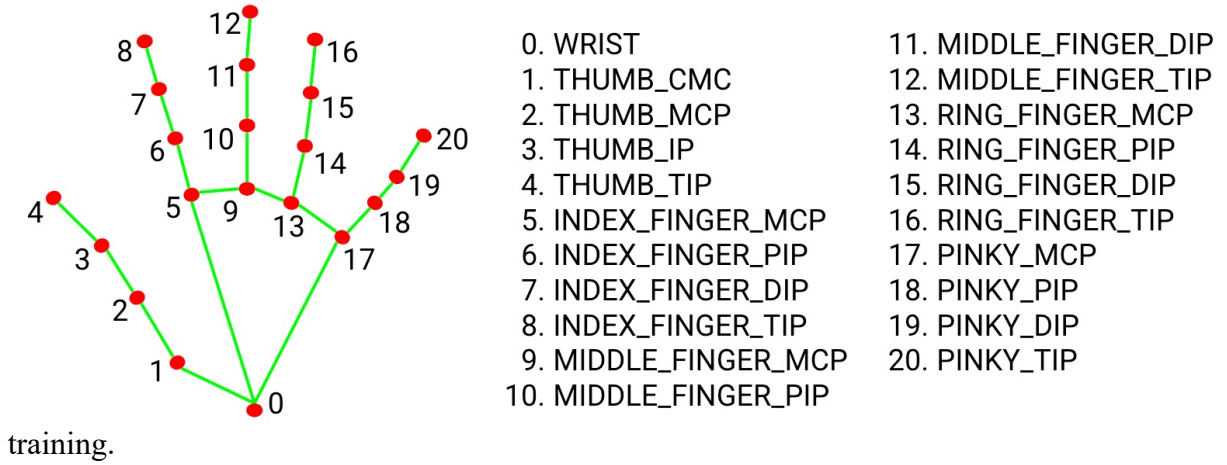


Figure 3.2 Labelled Keypoints

Figure 3.2: It represents a structured hand-keypoint model used for gesture analysis. It defines 21 anatomically meaningful landmarks, starting from the wrist and extending through the thumb, index, middle, ring, and pinky fingers. Each finger is described using joints such as CMC, MCP, PIP, DIP, and the fingertip, capturing precise spatial relationships. These landmarks collectively form a skeletal representation of the hand, enabling accurate feature extraction. Such a representation is fundamental for robust gesture recognition, as it preserves both finger articulation and overall hand posture in a compact, numerical form.

3.5 Feature Representation and Label Encoding

Each image is represented as a 63-dimensional feature vector derived from hand landmarks. Corresponding labels are assigned numerical values using label encoding to ensure compatibility with deep learning classifiers.

The dataset is split into training, validation, and testing sets in the ratio of 80:10:10 using stratified sampling to preserve class distribution.

3.6 Model Architecture

A deep neural network (DNN) is used for gesture classification. The architecture consists of:

- Input layer with 63 neurons
- Two fully connected hidden layers with ReLU activation
- Dropout layers for regularization
- Softmax output layer corresponding to the number of gesture classes

The model is trained using the Adam optimizer and sparse categorical cross-entropy loss function. Early stopping is employed to prevent overfitting.

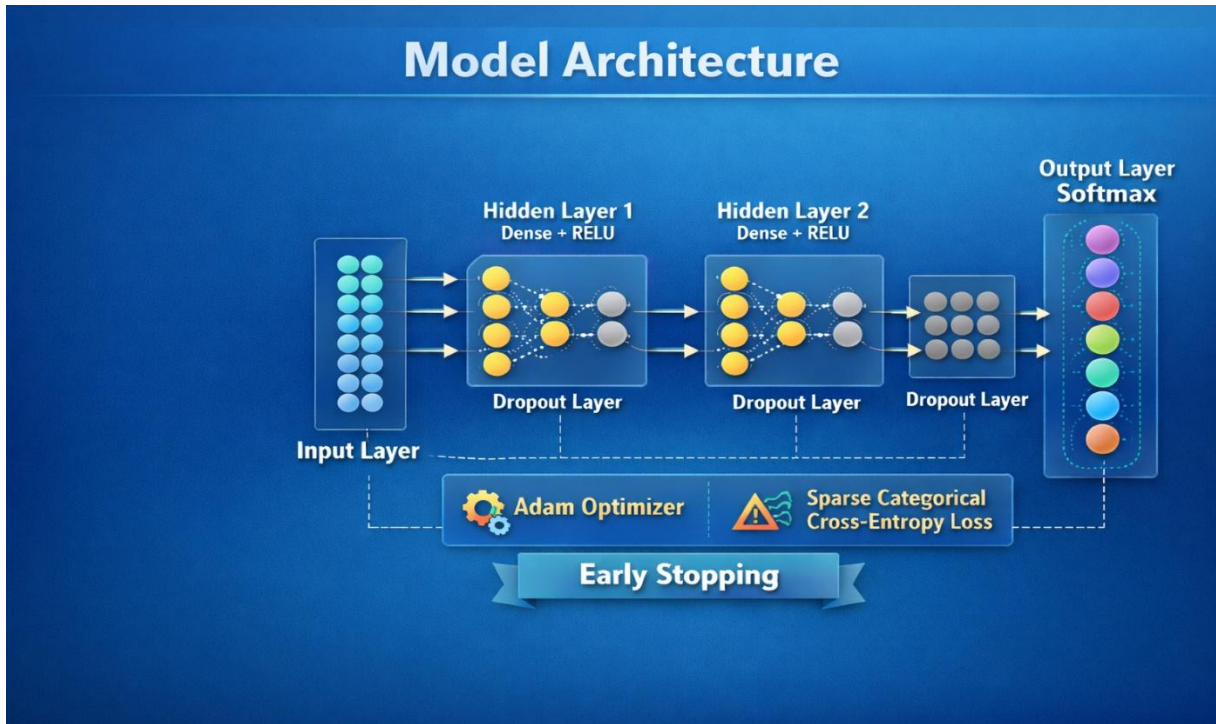


Figure 3.3: Model Architecture

Figure 3.3 : It illustrates a deep neural network architecture used for gesture classification. Input features are processed through two fully connected hidden layers with ReLU activation to learn meaningful patterns. Dropout layers are included to prevent overfitting and improve generalization. The final softmax output layer produces probability scores for each gesture class. The model is trained using the Adam optimizer with sparse categorical cross-entropy loss and employs early stopping to ensure efficient and stable training.

3.7 Transfer Learning Strategy

To study the effectiveness of transfer learning, a two-stage training strategy is implemented:

1. Base Model Training:

The model is first trained on ISL numbers (0–9) to learn fundamental hand shape

representations.

2. Transfer Learning for Alphabets:

The learned weights from the base model are reused, and higher layers are fine-tuned on the ISL alphabet dataset.

This approach aims to leverage shared hand gesture features between numbers and alphabets while reducing training time and data dependency.

3.8 Model Evaluation

Model performance is evaluated using multiple metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics provide a comprehensive assessment of classification performance and class-wise behavior. Visualization tools are used to generate confusion matrices and classification reports for analysis.

3.9 Real-Time Gesture Recognition and Translation

The trained model is integrated with a webcam-based interface for real-time gesture recognition. Live video frames are captured, and hand keypoints are extracted in real time using MediaPipe. The extracted features are passed to the trained classifier to predict the corresponding ISL gesture.

Predicted gestures are accumulated and displayed as text output, forming words or simple sentences. This real-time deployment demonstrates the practical applicability of the proposed system.

3.10 Summary

This chapter presented a comprehensive methodology for ISL recognition using keypoint-based deep learning. By combining efficient feature extraction, robust classification, and real-time deployment, the proposed approach addresses key limitations of existing systems. The inclusion of transfer learning further explores performance optimization under varying data conditions.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview

This chapter presents the experimental results obtained from the proposed Indian Sign Language (ISL) recognition system and provides a detailed discussion of model performance. The experiments were conducted using multiple training strategies, including training from scratch and transfer learning. Performance evaluation was carried out using standard classification metrics to assess both accuracy and reliability of the system.

4.2 Experimental Setup

The experiments were performed on a system equipped with a standard CPU and integrated GPU. The dataset comprised ISL hand gesture images representing numbers (0–9) and alphabets (A–Z). Hand keypoints were extracted using MediaPipe, resulting in a 63-dimensional feature vector per sample.

The dataset was split into training, validation, and testing subsets in the ratio of 80:10:10 using stratified sampling. The models were trained using the Adam optimizer with sparse categorical cross-entropy loss. Early stopping was applied to prevent overfitting and to ensure optimal generalization.

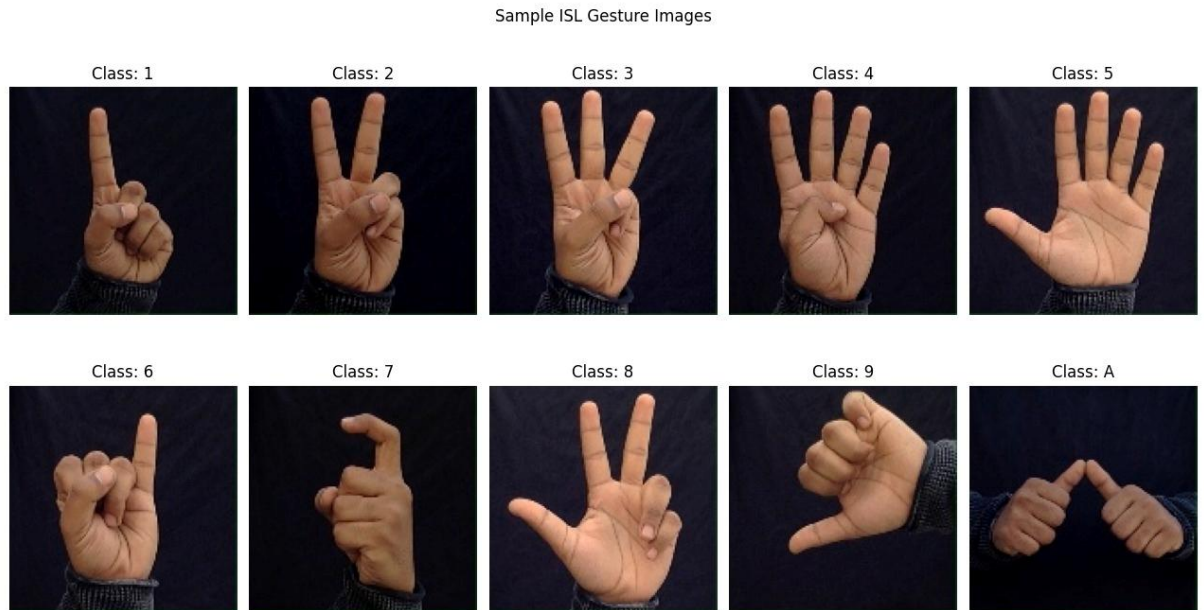


Figure 4.1 ISL hand gesture images representing numbers (0–9)

Figure 4.1 : The visual presents a set of sample Indian Sign Language gestures representing numerical classes and an alphabet gesture. Each hand pose corresponds to a distinct class label, demonstrating clear variations in finger positioning and hand orientation. The gestures progress from simple configurations, such as single or multiple raised fingers for numbers, to more complex shapes used for alphabet representation. These samples highlight the consistency within each class and the visual diversity across classes, which is essential for training and evaluating gesture recognition models effectively.

4.3 Performance of Initial Sentence-Level Dataset

Initial experiments were conducted using an ISL sentence-level video dataset. Due to the limited number of samples available after splitting into training and testing sets, the model was unable to learn discriminative features effectively. As a result, the achieved accuracy was approximately **41%**, indicating poor generalization.

This experiment highlighted the critical importance of dataset size and diversity in deep learning-based sign language recognition systems and motivated the transition to a larger and more structured dataset.

4.4 Results on Alphabet and Number Dataset (From Scratch)

The model trained from scratch using the combined ISL alphabet and number dataset demonstrated excellent performance. The test accuracy achieved was **99.93%**, indicating the effectiveness of keypoint-based feature representation and deep neural network classification.

The classification report showed high precision, recall, and F1-score across all gesture classes, with minimal misclassification. The confusion matrix further confirmed that the model was able to distinguish between visually similar gestures with high confidence.

The high accuracy can be attributed to:

- Sufficient dataset size
- Robust landmark-based feature extraction
- Effective regularization techniques

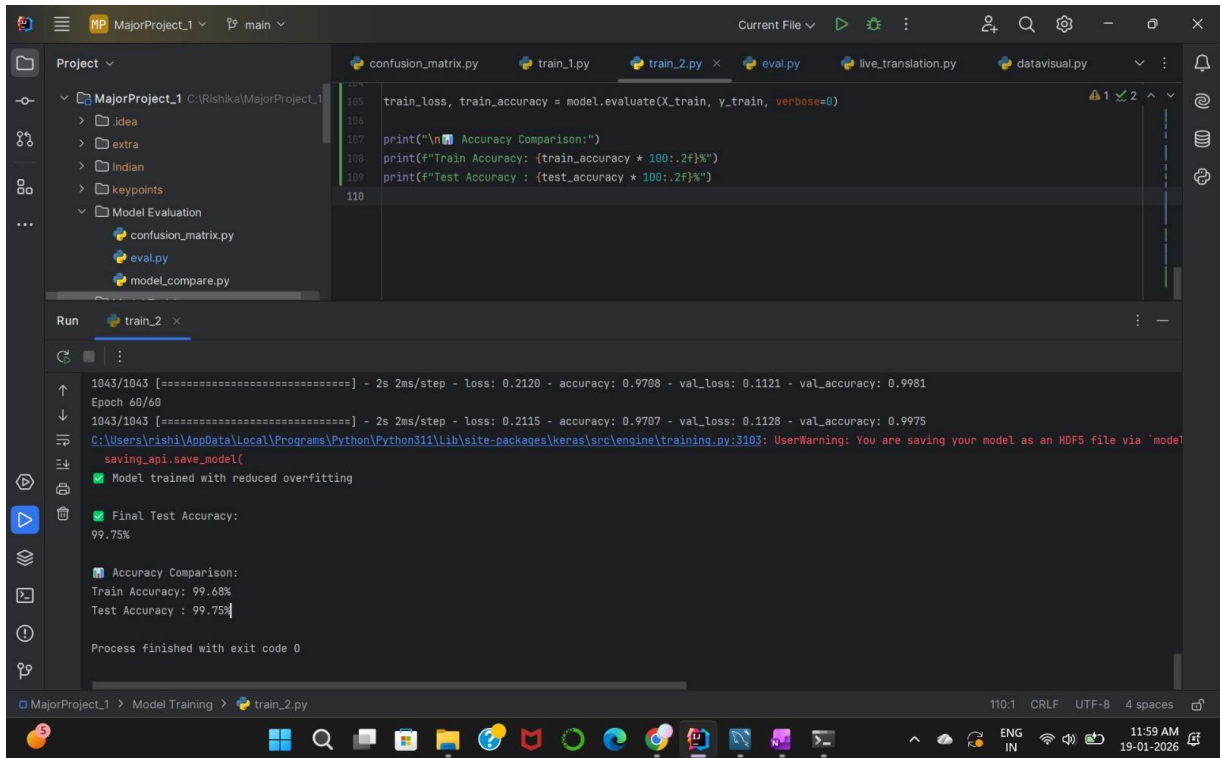


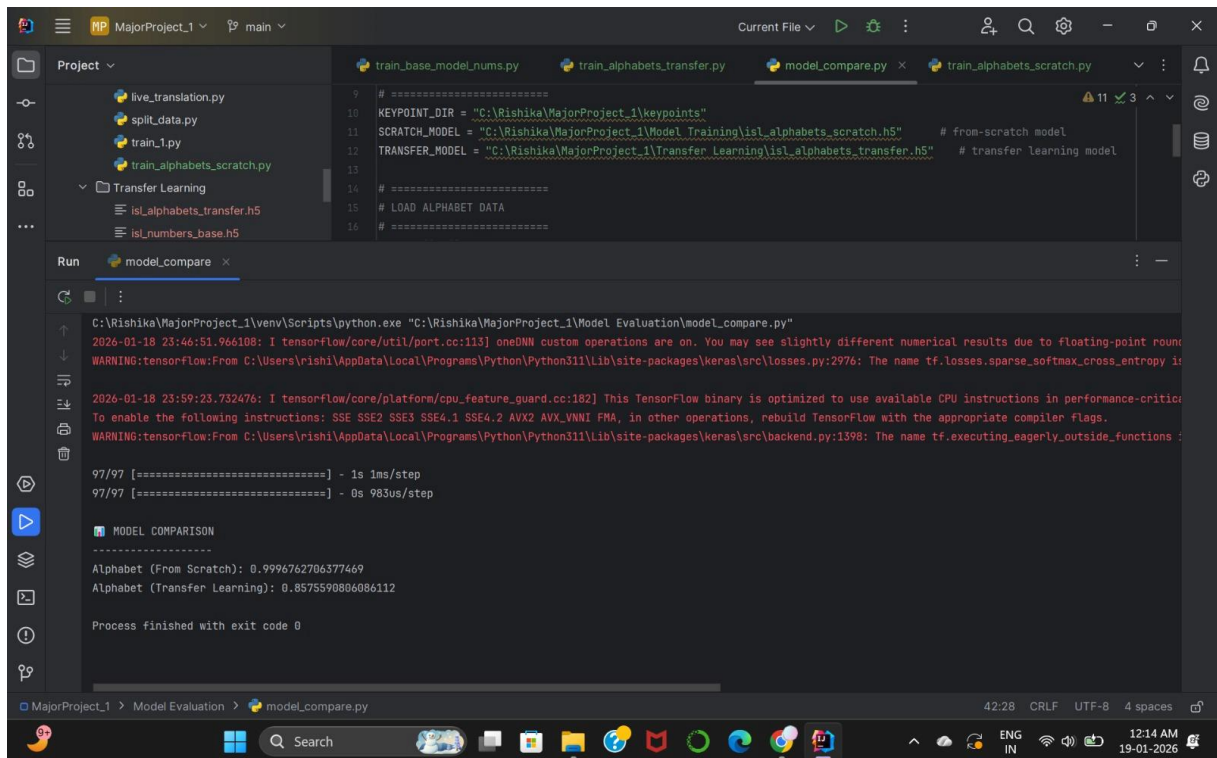
Figure 4.2 Train and Test accuracy

Figure 4.2 : The results indicate that the trained model achieves very high performance with minimal overfitting. The final training accuracy is approximately 99.68%, while the test accuracy reaches 99.75%, showing strong generalization on unseen data. The validation accuracy remains consistently high (around 99.8%) with low training and validation loss, confirming stable learning behavior. The message indicating reduced overfitting further supports that the model is well-optimized and reliable for gesture recognition tasks.

4.5 Transfer Learning Results

Transfer learning experiments were conducted by training a base model on ISL numbers and fine-tuning it on the alphabet dataset. While transfer learning reduced training time, the achieved accuracy was lower compared to the model trained from scratch.

This performance difference indicates that although numbers and alphabets share common hand features, alphabet gestures exhibit higher variability, requiring more specialized feature learning. In this scenario, training from scratch proved more effective due to the availability of sufficient training data.



```
train_base_model_nums.py train_alphabets_transfer.py model_compare.py train_alphabets_scratch.py
# =====
9 KEYPOINT_DIR = "C:\\Rishika\\MajorProject_1\\keypoints"
10
11 SCRATCH_MODEL = "C:\\Rishika\\MajorProject_1\\Model Training\\isl_alphabets_scratch.h5" # from-scratch model
12 TRANSFER_MODEL = "C:\\Rishika\\MajorProject_1\\Transfer Learning\\isl_alphabets_transfer.h5" # transfer learning model
13
14 # =====
15 # LOAD ALPHABET DATA
16 # =====
Run model_compare x
C:\\Rishika\\MajorProject_1\\venv\\Scripts\\python.exe "C:\\Rishika\\MajorProject_1\\Model Evaluation\\model_compare.py"
2026-01-18 23:46:51.966108: I tensorflow/core/util/port.cc:113] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round
WARNING:tensorflow:From C:\\Users\\rishi\\AppData\\Local\\Programs\\Python\\Python311\\Lib\\site-packages\\keras\\src\\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is
2026-01-18 23:59:23.732476: I tensorflow/core/platform/cpu_feature_guard.cc:182] This TensorFlow binary is optimized to use available CPU instructions in performance-critical
To enable the following instructions: SSE SSE2 SSE3 SSE4.1 SSE4.2 AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:From C:\\Users\\rishi\\AppData\\Local\\Programs\\Python\\Python311\\Lib\\site-packages\\keras\\src\\backend.py:1398: The name tf.executing_eagerly_outside_functions
97/97 [=====] - 1s 1ms/step
97/97 [=====] - 0s 983us/step
MODEL COMPARISON
-----
Alphabet (From Scratch): 0.9996762706377469
Alphabet (Transfer Learning): 0.8575590806086112
Process finished with exit code 0
MajorProject_1 > Model Evaluation > model_compare.py 42:28 CRLF UTF-8 4 spaces 12:14 AM 19-01-2026
```

Figure 4.3 Transfer Learning Results

Figure 4.3 : The results show a clear performance comparison between two alphabet recognition models. The model trained from scratch achieves a very high accuracy of approximately 99.96%, indicating excellent learning when trained directly on the alphabet dataset. In contrast, the transfer learning model attains an accuracy of about 85.75%, which is noticeably lower. This highlights that, for this task, training the alphabet model independently is more effective than transferring knowledge from a numbers-based model, likely due to differences in gesture patterns between the two domains.

4.6 Comparative Analysis

A comparative analysis between the different training strategies is summarized below:

- **Sentence-Level Dataset:** Low accuracy due to insufficient data
- **Alphabet and Number Dataset (From Scratch):** Highest accuracy and robustness
- **Transfer Learning Approach:** Faster convergence but reduced accuracy

This comparison demonstrates that transfer learning is most beneficial when data availability is limited, whereas training from scratch yields superior performance when large datasets are available.

Table 4.1 Comparative Analysis

Model	Dataset Used	Training Approach	Accuracy
Sentence-Level ISL Model	ISL Sentence Video Dataset	From Scratch	41%
ISL numbers+Alphabet model	Numbers(0-9) + Alphabets (A-Z)	From Scratch	~99%
ISL Alphabet model	Alphabets (A-Z)	From Scratch	~99.68%
ISL Alphabet model	Alphabets (A-Z)	Transfer Learning (numbers to Alphabets)	~85.7%

Table 4.1 : The table compares different ISL models based on the dataset used, training strategy, and achieved accuracy. The sentence-level ISL model is trained from scratch on an ISL sentence video dataset and achieves relatively lower accuracy due to the complexity of continuous gestures. In contrast, the ISL numbers and alphabet models trained from scratch show very high accuracy, indicating effective learning for simpler, well-defined gestures. The alphabet-only model achieves the highest performance when trained independently, while the transfer learning approach—using knowledge from the numbers model to train the alphabet model—shows moderate accuracy, reflecting partial knowledge reuse with some domain mismatch.

4.7 Real-Time Recognition Performance

The trained model was successfully deployed in a real-time environment using a webcam. The system was able to recognize ISL gestures with minimal latency and display the predicted output as text. Despite minor variations due to camera orientation and lighting conditions, the model maintained consistent performance.

The real-time implementation validates the practical usability of the proposed system for interactive ISL translation applications.

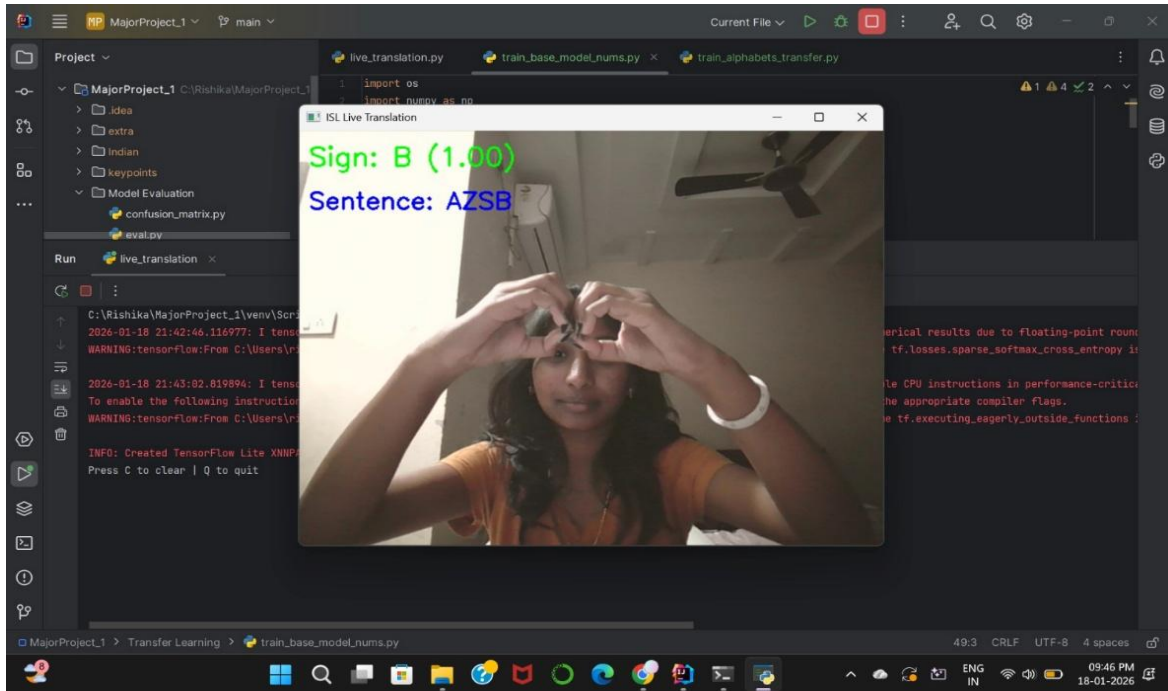


Figure 4.4 ISL Live Translation for alphabet B

Figure 4.4 : The image illustrates the real-time functioning of the ISL live translation system during alphabet recognition. The camera captures a hand gesture, which is correctly identified as the alphabet “B” with full confidence (1.00). The recognized gesture is immediately displayed on the screen and appended to the cumulative output, forming the sentence “AZSB”. This demonstrates continuous gesture detection, high prediction confidence, and accurate real-time translation of ISL alphabets into text.



Figure 4.5 ISL Live Translation for alphabet O

Figure 4.5 : The image demonstrates the real-time working of the ISL live translation system. A hand gesture is captured through a camera feed and processed instantly by the trained model. The system correctly recognizes the gesture as the alphabet “O” with full confidence (1.00), which is displayed on the screen. The detected sign is then appended to form a cumulative output, shown as a sentence (“CBZO”), illustrating continuous gesture recognition. This confirms the system’s ability to perform accurate, real-time ISL recognition and translation.

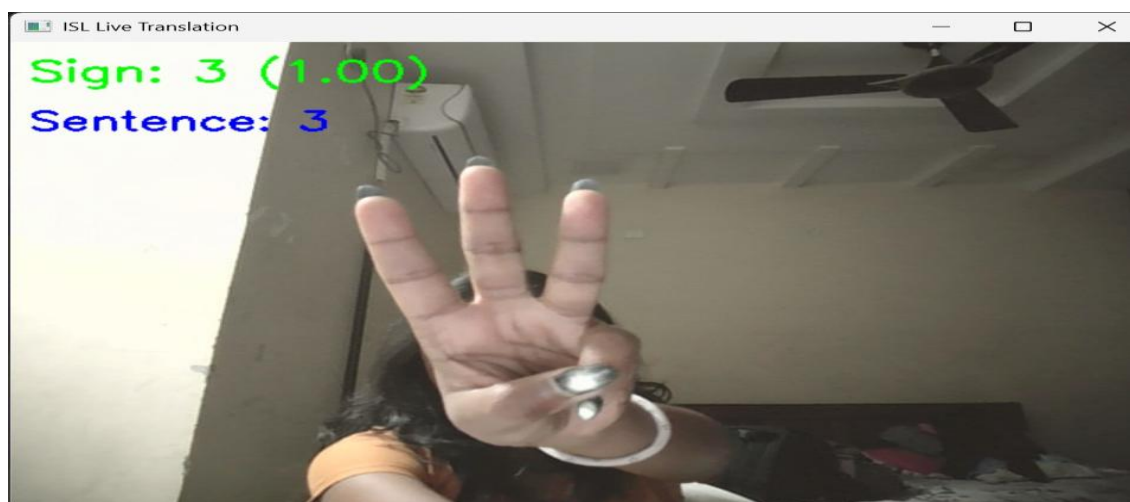


Figure 4.6 ISL Live Translation for Number 3

Figure 4.6 : The image shows the real-time operation of the ISL live translation system while recognizing a numerical gesture. The hand gesture representing the number “3” is captured through the camera and accurately identified by the model with full confidence (1.00). The recognized sign is displayed instantly and added to the output sentence, which in this case is “3”. This demonstrates the system’s effectiveness in detecting numerical ISL gestures and translating them correctly in real time.

4.8 Discussion

The experimental results clearly demonstrate the effectiveness of the proposed keypoint-based approach for ISL recognition. The significant improvement in accuracy compared to the initial dataset confirms the importance of dataset quality and feature representation. While transfer learning did not outperform training from scratch in this case, it remains a viable strategy for future extensions involving larger gesture

vocabularies or sentence-level recognition.

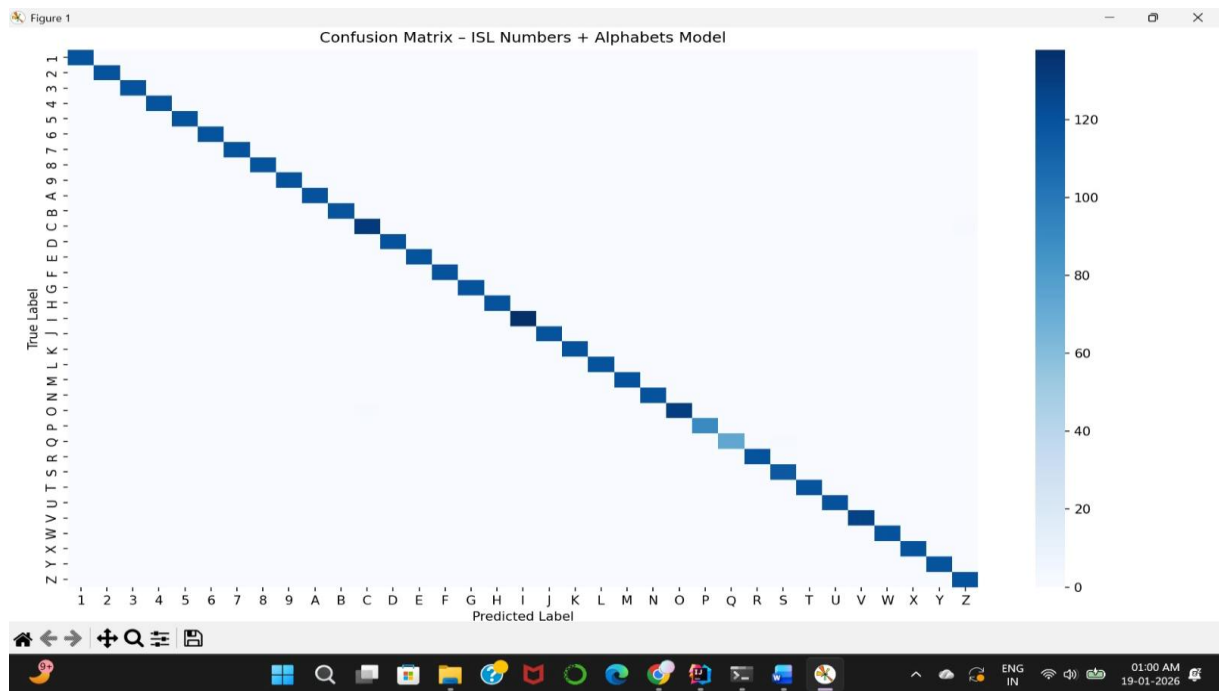


Figure 4.7 Confusion Matrix

Figure 4.7 : The image shows the confusion matrix for the ISL Numbers and Alphabets classification model. Most of the values are concentrated along the main diagonal, indicating that the model correctly predicts the true class for almost all numbers and alphabet gestures. The strong, dark diagonal pattern reflects very high classification accuracy and minimal misclassification. Only a few lighter off-diagonal cells appear, showing rare confusion between visually similar gestures. Overall, the matrix confirms that the model performs reliably and consistently across all ISL number and alphabet classes.

4.9 Summary

This chapter analyzed the experimental outcomes of the proposed ISL recognition system. The results indicate that the combination of MediaPipe-based keypoint extraction and deep learning classification provides a highly accurate and efficient solution. The system's ability to perform real-time gesture recognition further highlights its practical applicability.

CHAPTER 5

CONCLUSIONS AND FUTURE SCOPE

5.1 Conclusions

This project presented an effective and practical approach for Indian Sign Language (ISL) recognition and translation using hand keypoint extraction and deep learning techniques. The initial phase of the study involved experimenting with a sentence-level ISL video dataset; however, due to the limited dataset size and insufficient variation, the model achieved low recognition accuracy. This limitation highlighted the critical role of dataset quality and scale in deep learning-based sign language recognition systems.

To overcome this challenge, the project adopted a larger and more structured dataset consisting of ISL alphabets and numbers. By utilizing MediaPipe for hand landmark extraction, the system represented each gesture using a compact 63-dimensional feature vector, significantly reducing computational complexity while preserving essential spatial information. A deep neural network classifier was trained on these features, resulting in excellent classification performance with a test accuracy of approximately 99.9%.

The experimental results demonstrated that keypoint-based feature extraction is highly effective for ISL recognition, providing robustness against background noise, lighting variations, and camera inconsistencies. The comparison between training from scratch and transfer learning approaches revealed that while transfer learning offers faster convergence, training from scratch achieved superior accuracy when sufficient data was available. This finding emphasizes that the choice of training strategy should be guided by dataset size and diversity.

Furthermore, the trained model was successfully integrated into a real-time webcam-based system, enabling live gesture recognition and translation into text. The system exhibited low latency and reliable performance, validating its practical applicability for real-world communication assistance. Overall, the project achieved its objectives of developing an accurate, efficient, and real-time ISL recognition system.

5.1 Future Scope

Although the proposed system demonstrates high accuracy and real-time performance, several opportunities exist for further enhancement and research:

1. **Sentence-Level Sign Language Translation:**

Future work can extend the system from isolated gesture recognition to continuous sentence-level translation by incorporating temporal modeling techniques such as Long Short-Term Memory (LSTM) networks, Transformers, or Temporal Convolutional Networks.

2. **Integration of Facial and Body Keypoints:**

ISL communication involves facial expressions and upper-body movements in addition to hand gestures. Incorporating pose and facial landmarks can significantly improve semantic understanding and translation accuracy.

3. **Larger and More Diverse ISL Datasets:**

Expanding the dataset to include multiple signers, varying backgrounds, and different camera perspectives can enhance model generalization and robustness.

4. **Advanced Transfer Learning Strategies:**

Future research can explore pretraining on larger international sign language datasets and fine-tuning on ISL-specific data to improve performance in low-data scenarios.

5. **Mobile and Edge Deployment:**

Optimizing the model for deployment on mobile devices or edge platforms can increase accessibility and enable on-device real-time translation without dependency on high-performance hardware.

6. **Bidirectional Translation:**

The system can be extended to support text-to-sign generation using animated avatars, enabling complete two-way communication between hearing-impaired and non-hearing-impaired users.

7. **Explainability and Ethical Considerations:**

Future implementations can incorporate explainable AI techniques to improve model transparency and address ethical concerns related to accessibility and inclusivity.

5.3 Final Remarks

In conclusion, this project provides a strong foundation for Indian Sign Language recognition using efficient keypoint-based deep learning techniques. The results demonstrate both technical effectiveness and real-world feasibility. With further research and development, the proposed system has the potential to evolve into a comprehensive assistive communication platform, contributing meaningfully to social inclusion and accessibility for the hearing-impaired community.

REFERENCES

- [1] T. Starner and A. Pentland,
“Real-time American Sign Language recognition using desk and wearable computers,”
IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 20, no. 12, pp. 1371–1375, 1997.
- [2] S. C. W. Ong and S. Ranganath,
“Automatic sign language analysis: A survey,”
Pattern Recognition, vol. 38, no. 10, pp. 1659–1679, 2005.
- [3] L. Pigou, A. van den Oord, S. Dieleman, M. Van Herreweghe, and J. Dambre,
“Beyond Temporal Pooling: Recurrence and Temporal Convolutions for Gesture Recognition,”
International Journal of Computer Vision, vol. 126, no. 2–4, pp. 430–439, 2018.
Sfar AR, Chtourou Z, Natalizio E, and Challal Y. a road map for internet of things security issues. Digit Commun Network.
- [4] N. C. Camgoz, S. Hadfield, O. Koller, and R. Bowden,
“Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation,”
in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 10023–10033.
- [5] F. Zhang, S. Liu, J. Zhang, and Y. Chen,
“Skeleton-Based Hand Gesture Recognition Using MediaPipe,”
in *Proceedings of the International Conference on Artificial Intelligence and Computer Vision*, 2021.
- [6] C. Keskin, F. Kırac, Y. E. Kara, and L. Akarun,
“Hand Pose Estimation and Hand Shape Classification Using Multi-layered Randomized Decision Forests,”
in *European Conference on Computer Vision (ECCV)*, 2013, pp. 852–863.
- [7] N. Neverova, C. Wolf, G. Taylor, and F. Nebout,
“Multi-scale Deep Learning for Gesture Detection and Localization,”
in *European Conference on Computer Vision (ECCV)*, 2014, pp. 474–490.

- [8] P. Molchanov, S. Gupta, K. Kim, and J. Kautz,
“Hand Gesture Recognition with 3D Convolutional Neural Networks,”
in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2016, pp. 1–7.
- [9] O. Koller, S. Zargaran, H. Ney, and R. Bowden,
“Deep Sign: Hybrid CNN-HMM for Continuous Sign Language Recognition,”
in *Proceedings of the British Machine Vision Conference (BMVC)*, 2017.
- [10] K. Simonyan and A. Zisserman,
“Two-Stream Convolutional Networks for Action Recognition in Videos,”
in *Advances in Neural Information Processing Systems (NeurIPS)*, 2014, pp. 568–576.
- [11] F. Ronchetti, F. Quiroga, and J. C. Estrella,
“Sign Language Recognition Without Vision-Based Sensors,”
in *Proceedings of the International Conference on Computing and Communication Technologies*, 2016.
- [12] A. Abid, F. Hussain, and A. Zafar,
“Real-Time Sign Language Recognition Using MediaPipe and Deep Learning,”
in *Proceedings of the International Conference on Emerging Technologies*, 2019.
- [13] N. C. Camgoz, O. Koller, S. Hadfield, and R. Bowden,
“Neural Sign Language Translation,”
in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7784–7793.
- [14] J. Liu, G. Wang, and W. Hu,
“Skeleton-Based Action Recognition Using Graph Convolutional Networks,”
in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 1466–1475.
- [15] A. Shrestha and A. Mahmood,
“Review of Deep Learning Algorithms and Architectures,”
IEEE Access, vol. 7, pp. 53040–53065, 2019
- [16] K. He, X. Zhang, S. Ren, and J. Sun,
“Deep Residual Learning for Image Recognition,”
in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

- [17] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber,
“Connectionist Temporal Classification: Labelling Unsegmented Sequence Data
with Recurrent Neural Networks,”
*in Proceedings of the 23rd International Conference on Machine Learning
(ICML), 2006, pp. 369–376.*
- [18] H. Cooper, B. Holt, and R. Bowden,
“Sign Language Recognition,”
in Visual Analysis of Humans, Springer, London, 2011, pp. 539–562.
- [19] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh,
“OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields,”
*IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 43, no. 1, pp.
172–186, 2021.*
- [20] R. Rastgoo, K. Kiani, and S. Escalera,
“Hand Sign Language Recognition Using Deep Learning: An Overview,”
Artificial Intelligence Review, vol. 50, no. 4, pp. 519–539, 2018.